Notes: La Porta, Lopez-de-Silanes, Shleifer, and Vishny (JF, 2002)

Investor Protection and Corporate Valuation

Contents

1 Big picture

1.1 Research motivation

La Porta, Lopez-de-Silanes, Shleifer, and Vishny study whether investor protection is capitalized into firm value. More precisely, they analyze whether minority shareholders value firms more highly when the legal system constrains diversion by controlling shareholders, and whether controlling shareholders' own cash-flow ownership reduces the incentive to divert.

The central empirical object is Tobin's q, interpreted as the outside minority shareholder's valuation of corporate assets. The central institutional object is shareholder protection, proxied by legal origin and anti-director rights. The central ownership object is the cash-flow stake held by the controlling shareholder.

1.2 Setting

The sample contains 539 large nonfinancial firms from 27 relatively wealthy economies. The sample is restricted to firms with a controlling shareholder, defined as an ultimate owner controlling more than 10 percent of the votes. This design focuses attention on the agency problem that is most relevant outside the United States: not dispersed shareholders versus hired managers, but minority shareholders versus controlling shareholders.

The paper uses two country-level investor protection proxies. The first is a common-law dummy, motivated by prior evidence that common-law countries tend to protect minority shareholders more strongly. The second is the anti-director rights index, which counts specific legal rights that strengthen minority shareholder power.

1.3 Core finding

Firms are valued more highly in countries with stronger shareholder protection. Firms with higher cash-flow ownership by the controlling shareholder also tend to have higher valuations. These findings are consistent with a model in which legal protection and ownership incentives both reduce value-destroying diversion.

The paper's most important result is therefore not only that law matters for financial development, but that law matters because it affects the price outside investors are willing to pay for claims on corporate cash flows.

1.4 Economic takeaway

Corporate valuation depends on expected expropriation. When outside investors expect the controlling shareholder to divert private benefits, they discount the firm's public equity claim. Stronger investor protection raises valuation by making diversion more costly. Higher cash-flow ownership raises valuation by making the controller internalize more of the cost of diversion.

2 Incremental contribution of the paper

2.1 Relative to law and finance

Earlier law-and-finance work documented that countries with better investor protection have broader financial markets, more listed firms, larger equity markets, and better access to external finance. This paper studies the valuation channel directly. It asks whether stronger protection raises the value of existing firms, not just whether it expands the size of financial markets.

This is a key incremental step. If legal protection increases valuation, then outside investors' willingness to supply capital rises because the expected share of cash flows returned to them rises. The valuation channel therefore gives a microeconomic mechanism linking law to financial development.

2.2 Relative to ownership and agency papers

The Jensen and Meckling ownership-incentive logic is usually studied in the United States in the context of managerial ownership. This paper imports that logic into a cross-country setting where the central agency conflict is different. The controller often has voting control but only partial cash-flow rights. The paper shows that cash-flow ownership by controlling shareholders can still matter for valuation.

2.3 Relative to private benefits and tunneling

The paper does not directly observe diversion, tunneling, transfer pricing, or private benefits. Instead, it infers the economic importance of expropriation from market valuations. If countries with stronger investor protection and firms with stronger controller cash-flow incentives have higher q, then the market appears to price expected expropriation.

2.4 Relative to comparative corporate governance

The paper connects country-level law with firm-level ownership. It does not treat corporate governance as only a firm-level choice or only a country-level institution. Instead, law and ownership incentives interact: legal protection changes the cost of expropriation, while cash-flow ownership changes the controller's private incentive to expropriate.

3 Model

3.1 Players and primitives

A firm is controlled by a single entrepreneur or controlling shareholder. The controller has cash-flow ownership share α but effective control over corporate decisions. The firm invests cash I in a project that yields gross return R, so total pre-diversion profits are

$$RI.$$

The controller can divert a fraction s of profits before distributing the residual pro rata to shareholders.

Diversion is costly. If the controller diverts share s, he receives private benefits sRI, but pays a waste cost

$$c(k,s)RI,$$

where k measures the quality of shareholder protection. A higher k means stronger investor protection and makes diversion more costly.

The cost function satisfies

$$c_k > 0, \quad c_s > 0, \quad c_{ss} > 0, \quad c_{ks} > 0.$$

The final condition, $c_{ks} > 0$, is the crucial legal protection condition: better shareholder protection raises the marginal cost of diversion.

3.2 Controller's objective

The controller receives two components of payoff. First, he receives his cash-flow share of distributed profits:

$$\alpha(1-s)RI.$$

Second, he receives private diversion benefits net of the cost of diversion:

$$sRI - c(k,s)RI.$$

Thus, after dividing by RI, the controller maximizes

$$U(s) = \alpha(1-s) + s - c(k,s).$$

The first-order condition is

$$U_s = -\alpha + 1 - c_s(k,s) = 0,$$

or equivalently

$$c_s(k,s^*) = 1 - \alpha.$$

This is the Jensen-Meckling incentive condition in a controlling-shareholder environment. The marginal benefit of diversion is $1-\alpha$, because diversion transfers value from outside shareholders to the controller. The marginal cost is $c_s(k,s)$.

3.3 Effect of legal protection on diversion

Differentiate the first-order condition with respect to k:

$$c_{ks}(k,s^*) + c_{ss}(k,s^*)\frac{ds^*}{dk} = 0.$$

Hence

$$\frac{ds^*}{dk} = -\frac{c_{ks}(k,s^*)}{c_{ss}(k,s^*)} < 0.$$

Because $c_{ks} > 0$ and $c_{ss} > 0$, stronger shareholder protection reduces equilibrium diversion.

Result: legal protection lowers expropriation.

3.4 Effect of cash-flow ownership on diversion

Differentiate the first-order condition with respect to α:

$$c_{ss}(k, s^*)\frac{ds^*}{d\alpha} = -1.$$

Thus

$$\frac{ds^*}{d\alpha} = -\frac{1}{c_{ss}(k, s^*)} < 0.$$

A higher cash-flow stake makes diversion less attractive because the controller bears a larger share of the lost pro rata distribution.

Result: higher controller cash-flow ownership lowers expropriation.

3.5 Minority-shareholder valuation

For minority outside shareholders, the firm value reflects only the distributed cash flows, not the private benefits of control. Thus the model's Tobin's q is

$$q = (1 - s^*)R.$$

The comparative statics are immediate:

$$\frac{dq}{dk} = -R\frac{ds^*}{dk} > 0,$$

$$\frac{dq}{d\alpha} = -R\frac{ds^*}{d\alpha} > 0,$$

and

$$\frac{dq}{dR} = 1 - s^* > 0.$$

Thus the model yields three baseline predictions:

1. Firms in more protective legal regimes should have higher Tobin's q.
2. Firms with higher controller cash-flow ownership should have higher Tobin's q.
3. Firms with better investment opportunities should have higher Tobin's q.

3.6 Interaction between law and cash-flow ownership

The paper also asks whether ownership incentives matter less when legal protection is strong. Intuitively, if law already makes expropriation costly, the marginal value of controller cash-flow ownership may be smaller.

The general model does not sign this cross-partial without further restrictions because it depends on third derivatives of the cost-of-theft function. With the quadratic cost function

$$c(k, s) = \frac{1}{2}ks^2,$$

we have

$$c_s(k, s) = ks.$$

The first-order condition becomes

$$ks^* = 1 - \alpha,$$

so

$$s^* = \frac{1-\alpha}{k}.$$

Then

$$q = \left(1 - \frac{1-\alpha}{k}\right) R.$$

The marginal effect of cash-flow ownership is

$$\frac{dq}{d\alpha} = \frac{R}{k}.$$

Differentiating again with respect to k,

$$\frac{d^2 q}{d\alpha\, dk} = -\frac{R}{k^2} < 0.$$

Thus, under the quadratic cost-of-theft function, law and ownership incentives are substitutes: cash-flow ownership matters more for valuation in weak investor protection regimes.

4 Empirical design

4.1 Sample construction

The sample includes the largest 20 firms by market capitalization in each of 27 countries, subject to the requirement that the firm has a controlling shareholder with more than 10 percent of voting rights. The final sample has 539 firms.

The authors exclude financial firms because valuation ratios for financial firms are not comparable to those for nonfinancial firms. They also exclude affiliates of foreign firms, because the relevant governance system for these affiliates may not be the local country's governance system.

4.2 Why focus on firms with controlling shareholders

The empirical strategy is designed to keep the controller's power relatively constant. If every sample firm has a controlling shareholder, variation in valuation can be interpreted more naturally as coming from differences in legal protection and controller cash-flow incentives rather than from the mere presence or absence of control.

This choice also fits the international setting. In most countries, large publicly traded firms are not widely held. They are usually controlled by families, business groups, the state, or another corporation.

4.3 Control rights

A shareholder is treated as controlling a firm if his direct and indirect voting rights exceed 10 percent. Indirect control is measured through control chains. If the controlling shareholder controls firm B, and firm B controls x percent of firm A, then the shareholder is treated as controlling x percent of the votes of firm A.

When more than one shareholder has more than 10 percent voting control, the paper selects the ultimate owner with the highest minimum voting stake along the control chain.

4.4 Cash-flow rights

Cash-flow rights measure the fraction of the firm's ultimate cash-flow claims owned by the controlling shareholder. If a controller owns fraction y of firm B, and firm B owns fraction x of the sample firm, the controller's cash-flow ownership along that chain is xy. If there are multiple chains, the cash-flow ownership shares are added across chains.

This distinction is central because many controllers use pyramids, dual-class shares, or cross-holdings to maintain voting control with lower cash-flow ownership.

4.5 Investor protection proxies

The paper uses two proxies for shareholder protection:

- **Common law:** equals one when the country's company law or commercial code has common-law origin. Prior law-and-finance work shows that common-law countries generally offer stronger shareholder protection.
- **Anti-director rights:** an index from zero to six measuring specific legal rights of minority shareholders, including voting by mail, no share-blocking before meetings, cumulative voting or proportional representation, oppression remedies, preemptive rights, and the threshold for calling extraordinary meetings.

4.6 Valuation measures

The main valuation measure is Tobin's q:

$$q = \frac{\text{market value of assets}}{\text{book value of assets}}.$$

The numerator is book assets minus book equity and deferred taxes plus the market value of common equity. The denominator is book assets, used as a proxy for replacement value.

The paper also uses industry-adjusted q:

$$q_i - \text{world median } q_{\text{industry}(i)}.$$

The purpose is to remove differences in valuation that reflect global industry composition rather than investor protection or ownership incentives.

4.7 Investment opportunities

The model predicts that R, the return on investment opportunities, raises q. The empirical proxy is sales growth, measured as the three-year geometric average annual growth rate in sales. For industry-adjusted regressions, the authors use industry-adjusted sales growth.

5 Preliminary evidence

5.1 Legal origin and valuation

The preliminary country-level comparison shows that common-law countries have stronger anti-director rights and higher median valuations. The median anti-director rights score is four for common-law countries and two for civil-law countries. The median of country median Tobin's q is 1.3724 in common-law countries and 1.2022 in civil-law countries.

The difference in Tobin's q is statistically significant at the 5 percent level in the median test. This provides simple evidence that better legal protection is associated with higher valuation.

5.2 Why preliminary evidence is not enough

The simple comparison is informative but incomplete. Common-law countries may have better growth opportunities, different industry compositions, larger firms, or more liquid markets. The model also predicts that cash-flow ownership by the controller should matter. The regression analysis therefore adds sales growth, ownership, interactions, and industry adjustment.

6 Main regression evidence

6.1 Random-effects specification

The paper estimates random-effects regressions because the key legal variables vary at the country level. Country fixed effects would absorb legal origin and anti-director rights. Random effects allow the authors to exploit country-level variation while adjusting standard errors for within-country correlation.

The baseline specification can be written as

$$q_{ic} = \beta_0 + \beta_1 \text{Growth}_{ic} + \beta_2 \text{Protection}_c + \beta_3 \text{CF Rights}_{ic} + \beta_4 (\text{CF Rights}_{ic} \times \text{Protection}_c) + u_c + \varepsilon_{ic},$$

where u_c is a country random effect.

6.2 Raw Tobin's q results

Table III reports the raw q regressions. Growth in sales is positive and highly significant in all specifications, consistent with the model's investment-opportunity prediction.

Common law is positively associated with valuation. When common law is included by itself with growth, the coefficient is 0.2441 and is significant at the 10 percent level. When cash-flow rights and the interaction term are added, the common-law coefficient rises to 0.2848 and is significant at the 5 percent level.

The anti-director rights index is also positive. It is insignificant when included alone, but when cash-flow rights and the interaction are included, the coefficient is 0.1083 and significant at the 5 percent level. A two-point increase in the anti-director index, roughly the difference between civil-law and common-law medians, corresponds to about a 0.2 increase in Tobin's q.

Cash-flow rights are also positive. In the common-law specification, the coefficient is 0.2552 and significant at the 10 percent level. In the anti-director specification, the coefficient is 0.5228 and significant at the 5 percent level. The interaction coefficients are negative, as the model's substitute interpretation suggests, but not statistically significant.

6.3 Industry-adjusted results

Table IV repeats the exercise with industry-adjusted Tobin's q. The results remain similar. Common law is positive and significant; anti-director rights become positive and significant when combined with cash-flow rights; and cash-flow rights remain positively related to valuation.

The interpretation is important: the basic results do not appear to be driven by common-law countries having firms in globally high-valuation industries. The results survive after subtracting world industry median valuation.

6.4 Demeaned ownership results

A concern is that cash-flow ownership may be endogenous to the legal environment. For instance, controlling shareholders may choose lower cash-flow stakes in countries where legal protection is stronger. To address this, Table V replaces cash-flow rights with deviations from the country mean of cash-flow rights.

This approach uses only within-country variation in cash-flow ownership to identify the incentive effect. The results are weaker but still supportive. The incentive effect remains present at about the 10 percent significance level, and the investor-protection results weaken but do not disappear.

The key takeaway is that the positive ownership-valuation relation is not driven entirely by cross-country differences in average ownership structures.

6.5 Ownership, control, and wedge

Table VI shows country averages of cash-flow rights, control rights, and the wedge between control and cash-flow rights. The worldwide mean wedge is about 10 percentage points. The wedge is 11 percent in civil-law countries and 8 percent in common-law countries, with no statistically significant difference.

This table highlights a measurement challenge. Cash-flow rights and control rights are correlated. If higher cash-flow rights also imply higher control, then the positive incentive effect could be partly offset by a control-entrenchment effect. The relatively small wedge in the sample helps, but it does not fully solve the identification problem.

7 Robustness and alternative interpretations

7.1 Liquidity and capital-market size

One alternative explanation is that low valuation in weak investor protection countries reflects illiquidity rather than expropriation. The paper argues that a liquidity premium can lower price-cash-flow ratios, but it cannot easily explain Tobin's q below that of comparable firms because firms should invest until marginal q equals one in a no-agency-cost world.

The paper also checks firms with American Depositary Receipts. The ADR evidence does not support the view that liquidity is the primary driver: the valuation benefit of having an ADR is not concentrated in the less liquid civil-law markets where the liquidity story would predict the largest effect.

7.2 Investment opportunities

Past sales growth may be an imperfect proxy for growth opportunities. The paper tries alternative proxies, including capital expenditures to sales, asset growth, and future sales growth. The main conclusions are robust to these alternatives.

7.3 Firm size and sample selection

Common-law firms may be larger, and larger firms may have different valuations. The authors control for log sales and also broaden the sample by adding widely held firms and medium-size firms. The results remain broadly robust, and size itself enters with a negative coefficient in some specifications, which is inconsistent with a simple large-firm valuation explanation.

7.4 Multiple large shareholders

Another concern is that firms with multiple large shareholders have governance dynamics that differ from the one-controller model. The authors reestimate the analysis on firms with only one shareholder above the 10 percent threshold. The results hold, and the incentive effect is even larger in this subsample.

7.5 Endogeneity of ownership

Ownership is not randomly assigned. The controller's cash-flow rights may be chosen in response to legal protection, firm history, financing needs, or expected private benefits. The paper's main response is that ownership is historically sticky, especially outside the United States. The demeaned-ownership regressions are also used to focus on within-country variation.

This is not a complete solution. The ownership coefficient should be interpreted as evidence consistent with incentives, not as a fully clean causal estimate of cash-flow ownership.

8 Interpretation of the coefficients

8.1 Legal-protection effect

The legal-protection coefficients are economically meaningful. A common-law coefficient around 0.25 to 0.28 means that, holding sales growth and ownership constant, firms in common-law countries have Tobin's q roughly one-quarter point higher than comparable firms in civil-law countries.

Because the sample median q is close to 1.27, this is a substantial valuation difference. It suggests that expected expropriation can account for a meaningful portion of international valuation differences.

8.2 Cash-flow ownership effect

The cash-flow ownership coefficients are smaller in economic terms but consistent with the model. The paper interprets the raw-data coefficient of 0.255 as implying that increasing controller cash-flow ownership from 20 percent to 30 percent raises Tobin's q by roughly 0.026 in civil-law countries and by roughly 0.016 in common-law countries.

For the anti-director specification, the cash-flow rights coefficient implies a somewhat larger effect: increasing cash-flow ownership from 20 percent to 30 percent raises q by roughly 0.05 when the anti-director score is two and by roughly 0.03 when it is four.

8.3 Interaction effect

The interactions between cash-flow rights and investor protection are negative, consistent with the model's prediction that ownership incentives matter less when legal protection is strong. However, the interactions are generally not statistically significant. Thus the evidence for substitutability between law and ownership is suggestive but not definitive.

9 What the paper teaches

9.1 Investor protection affects prices

The paper's main lesson is that corporate governance institutions are priced. Outside investors pay more for the same corporate assets when the legal environment reduces expropriation risk.

9.2 Control is not enough; cash-flow rights matter

A controlling shareholder's voting power creates the ability to divert. Cash-flow ownership determines the incentive to refrain from diversion. The paper shows that this incentive channel matters even in a sample where all firms have controlling shareholders.

9.3 The relevant agency conflict differs across countries

In dispersed-ownership systems, the canonical agency problem is managers versus shareholders. In many countries, the more relevant problem is controlling shareholders versus minority shareholders. This paper shifts the valuation analysis to that latter problem.

9.4 Law and ownership are governance substitutes

The model suggests that legal protection and cash-flow ownership can substitute for each other. Strong law raises the marginal cost of diversion; high cash-flow rights lower the marginal benefit of diversion. The empirical evidence is strongest for their separate effects and weaker for their interaction.

10 Limitations and cautions

10.1 Legal origin is broad

Legal origin is a powerful proxy, but it bundles many institutional features: judicial philosophy, enforcement, political economy, disclosure, market development, and investor culture. The paper also uses anti-director rights, but country-level identification remains broad.

10.2 Tobin's q is imperfect

Tobin's q uses book assets as a proxy for replacement value. Accounting standards vary across countries. Industry adjustment helps, and the paper discusses consolidation rules, but measurement differences remain a concern.

10.3 Large-firm sample

The sample consists of large firms. These firms may have better disclosure, more analyst coverage, foreign listings, and reputational constraints than smaller firms. If anything, this may make it harder to detect the value of investor protection, but it also means the results may not generalize mechanically to smaller firms.

10.4 Ownership endogeneity

Cash-flow ownership may be endogenous. Demeaning by country helps isolate within-country variation, but the paper does not have a fully exogenous ownership instrument. The ownership results should therefore be interpreted more cautiously than the legal-protection results.

11 Connection to the broader literature

11.1 La Porta et al. law-and-finance sequence

This paper fits into the law-and-finance sequence: investor protection affects financial development, ownership concentration, dividend policy, and valuation. Here the key outcome is firm value. The paper therefore ties the legal system to the price of corporate claims.

11.2 Private benefits of control

The paper is closely related to studies of voting premia and private benefits of control. In countries where minority shareholders are poorly protected, control rights are more valuable because they allow more diversion. This paper studies the complementary implication: outside equity claims are less valuable.

11.3 Corporate ownership and valuation

The paper complements U.S. studies that find managerial ownership has both incentive and entrenchment effects. In this international setting, the cash-flow rights of controlling shareholders appear to have a positive incentive effect, while the separate effect of voting control is harder to identify.

12 A compact reading of the model and evidence

12.1 Model in one line

The controller chooses diversion until

$$\text{marginal cost of diversion} = \text{marginal private benefit of diversion},$$

that is,

$$c_s(k, s^*) = 1 - \alpha.$$

Better law raises the left-hand side; higher cash-flow ownership lowers the right-hand side. Both reduce s^*, and therefore raise

$$q = (1 - s^*)R.$$

12.2 Empirics in one line

Across 539 controlled firms in 27 countries, firms have higher valuations when they are located in countries with better shareholder protection and when controllers have higher cash-flow ownership.

12.3 The central economic intuition

Outside shareholders value a claim according to what they expect to receive, not according to total corporate cash flows. If controllers can tunnel cash flows away, outside equity is worth less. Law and cash-flow ownership are mechanisms that make tunneling less attractive.

13 Extracted figures and tables from the paper

13.1 Table I: The Variables

Table I

The Variables

This table describes the variables collected for the 27 countries included in our study. We present the description and the sources from which ea variable is collected.

Variable	Description
Common law	Equals one if the origin of the Company Law or Commercial Code of the country is the English Comm Law, and zero otherwise. Source: La Porta et al. (1998).
Civil law	Equals one if the Company Law or Commercial Code of the country originates in Roman Law, and ze otherwise. Source: La Porta et al. (1998).
Anti-director rights	Formed by adding one when: (1) the country allows shareholders to mail their proxy vote, (2) shareholde are not required to deposit their shares prior to the General Shareholders' Meeting, (3) cumulative voting proportional representation of minorities on the board of directors is allowed, (4) an oppressed minoriti mechanism is in place, (5) the minimum percentage of share capital that entitles a shareholder to call for Extraordinary Shareholders' Meeting is less than or equal to 10 percent (the sample median), or (6) wh shareholders have preemptive rights that can only be waived by a shareholders meeting. The range for t index is from zero to six. Source: La Porta et al. (1998).
Tobin's q	The ratio of the market value of assets to their replacement value at the end of the most recent fiscal yea The market value of assets is proxied by the book value of assets minus the book value of equity min deferred taxes plus the market value of common stock. The replacement value of assets is proxied by t book value of assets. Source: WorldScope (1997).
Industry-adjusted Tobin's q	Industry-adjusted Tobin's q is computed as the difference between Tobin's q and the *world median* Tobin's for the firm's industry. Industry control groups are defined at the three-digit SIC level whenever there a at least five *WorldScope* firms (excluding sample firms) in that group and at the two-digit SIC level othe

Table I: The Variables, part 1.

Growth in sales (GS)	Geometric average annual percentage growth in lagged (net) sales for up to three years depending on d availability. Sales are expressed in (US$) dollars. Source: WorldScope (1997).
Industry-adjusted GS	Average annual industry-adjusted growth in lagged (net) sales for up to three years depending on data av ability. Industry-adjusted GS is computed as the difference between GS and the *world median* GS for the fir industry. Industry control groups are defined at the three-digit SIC level whenever there are at least f *WorldScope* firms (excluding sample firms) in that group, and at the two-digit SIC level otherwise. Sou WorldScope (1997).
Control rights	Fraction of the firm's voting rights, if any, owned by its controlling shareholder. To measure control, we comb a shareholder's *direct* (i.e., through shares registered in her name) and *indirect* (i.e., through shares held entities that, in turn, she controls) *voting* rights in the firm. A shareholder has an *x percent indirect control* o firm A if: (1) she controls directly firm B which, in turn, directly controls x percent of the votes in firm A; (2) she controls directly firm C which in turn controls firm B (or a sequence of firms leading to firm B, each which has control over the next one, i.e., they form a control chain) which, in turn, directly controls x perc of the votes in firm A. A group of n companies form a *chain of control* if each firm 1 through $n-1$ controls consecutive firm. A firm in our sample has a controlling shareholder if the sum of her direct and indirect vot rights exceeds 10 percent. When two or more shareholders meet our criteria for control, we assign control to shareholder with the largest (direct plus indirect) voting stake.
CF rights	Fraction of the firm's ultimate cash-flow rights, if any, owned by its controlling shareholder. CF rights computed as the product of all the equity stakes along the control chain (see description of control rights for explanation of "control chains").
Wedge	The difference between control rights and cash flow rights.

Table I: The Variables, part 2.

Table I defines legal-origin variables, anti-director rights, valuation measures, growth variables, control rights, cash-flow rights, and the wedge.

13.2 Table II: Data; Table III: Random-Effects Regressions for Raw Data

Table II
Data

Panel A classifies countries by legal origin and presents medians by country for both the sample of 539 firms that have a controlling shareholder. Panel B reports tests of medians for civil versus common law legal origin. Variables are defined in Table I.

Country	Anti-director Rights	Tobin's q	Growth in Sales (%)
	Panel A: Medians		
Argentina	4	1.1494	14.07
Austria	2	1.1180	8.52
Belgium	0	1.2226	8.89
Denmark	2	1.5040	10.71
Finland	3	1.1018	15.25
France	3	1.2728	8.31
Germany	1	1.1921	7.31
Greece	2	1.6734	22.05
Italy	1	1.0320	7.00
Japan	4	1.3263	0.73
Korea	2	1.0725	20.47
Mexico	1	1.6388	−4.00
Netherlands	2	1.7404	12.88
Norway	4	1.1443	14.16
Portugal	3	1.0910	20.20
Spain	4	1.1560	5.05
Sweden	3	1.2123	16.20
Switzerland	2	1.3351	11.36
Civil law median	**2**	**1.2022**	**11.03**
Australia	4	1.3724	14.71
Canada	5	1.7533	17.38
Hong Kong	5	1.1626	10.64
Ireland	4	1.2937	12.64
Israel	3	1.1669	12.88
New Zealand	4	1.3344	17.15
Singapore	4	1.5486	25.70
United Kingdom	5	1.7165	10.22
United States	5	3.0815	9.94
Common law median	**4**	**1.3724**	**12.88**
Sample median	**3**	**1.2728**	**12.64**
	Panel B: Test of Medians (z-statistic)		
Civil versus common law	−3.53*	−2.16**	−1.29

*Significant at 1 percent level.

Table II: Data.

Table III
Random-Effects Regressions for Raw Data

The table presents results of random-effect regressions for the sample of 539 firms with a controlling shareholder. The dependent variable is Tobin's q. The independent variables are: (1) growth in sales, the three-year geometric average annual growth rate in sales; (2) common law, a dummy variable that equals one if the legal origin of the Company Law or Commercial Code of the country in which the firm is incorporated is Common Law and zero otherwise; (3) anti-director rights, the index of anti-director rights of the country in which the firm is incorporated; (4) CF Rights, the fraction of the cash-flow rights held by the firm's controlling shareholder; (5) the interaction between CF rights and common law; and (6) the interaction between CF rights and anti-director rights. Table I provides definitions for the variables. Standard errors are shown in parentheses.

	(1)	(2)	(3)	(4)
Constant	1.2986*	1.2192*	1.1559*	0.9480*
	(0.0836)	(0.0900)	(0.1649)	(0.1635)
Growth in sales	0.8275*	0.8191*	0.8314*	0.8258*
	(0.1403)	(0.1408)	(0.1403)	(0.1411)
Common law	0.2441***	0.2848**		
	(0.1400)	(0.1472)		
Anti-director rights			0.0735	0.1083**
			(0.0490)	(0.0478)
CF rights		0.2552***		0.5228**
		(0.1334)		(0.2680)
CF rights * common law		−0.0946		
		(0.2367)		
CF rights * anti-director rights				−0.1023
				(0.0828)
Overall R^2	0.0735	0.0771	0.0654	0.0759

*Significant at 1 percent level.
**Significant at 5 percent level.
***Significant at 10 percent level.

Table III: Random-Effects Regressions for Raw Data.

Table II compares medians across countries and legal origins; Table III estimates raw Tobin's q regressions with legal protection and cash-flow rights.

13.3 Table IV: Random-Effects Regressions for Industry-Adjusted Data; Table V: Random Effects and Demeaned Ownership

Table IV

Random-Effects Regressions for Industry-Adjusted Data

The table presents results of random-effects regressions for the sample of 539 firms with a controlling shareholder. The dependent variable is industry-adjusted Tobin's q. The independent variables are: (1) industry-adjusted growth in sales, the three-year geometric average annual growth rate in industry-adjusted sales; (2) common law, a dummy variable that equals one if the legal origin of the Company Law or Commercial Code of the country in which the firm is incorporated is Common Law and zero otherwise; (3) anti-director rights, the index of anti-director rights of the country in which the firm is incorporated; (4) CF rights, the fraction of the cash flow rights held by the firm's controlling shareholder; (5) the interaction between CF rights and common law; and (6) the interaction between CF rights and anti-director rights. Table I provides definitions for the variables. Standard errors are shown in parentheses.

	(1)	(2)	(3)	(4)
Constant	0.0705	−0.0181	−0.0639	−0.1906
	(0.0704)	(0.0758)	(0.1389)	(0.1318)
Industry-adjusted growth in sales	0.7505*	0.7487*	0.7534*	0.7443*
	(0.1336)	(0.1337)	(0.1336)	(0.1336)
Common law	0.1982***	0.2613**		
	(0.1199)	(0.1196)		
Anti-director rights			0.0659	0.0857**
			(0.0415)	(0.0377)
CF rights		0.2811**		0.4066*
		(0.1158)		(0.1583)
CF rights * common law		−0.2571		
		(0.2058)		
CF rights * anti-director rights				−0.0764
				(0.0481)
Overall R^2	0.0659	0.0801	0.0638	0.0851

*Significant at 1 percent level.
**Significant at 5 percent level.
***Significant at 10 percent level.

Table IV: Random-Effects Regressions for Industry-Adjusted Data.

Table V

Random Effects and Demeaned Ownership

The table presents random-effects regressions for the cross section of 539 firms in 27 countries that have a controlling shareholder. The depen dent variables are: (1) Tobin's q in Panel A; and (2) industry-adjusted Tobin's q in Panel B. The independent variables are: (1) growth in sales the three-year geometric average annual growth rate in sales; (2) industry-adjusted GS, the three-year geometric average annual growth rat in industry-adjusted sales; (3) common law, a dummy variable that equals one if the legal origin of the Company Law or Commercial Code o the country in which the firm is incorporated is common law and zero otherwise; (4) anti-director rights, the index of anti-director rights of th country in which the firm is incorporated; (5) CF rights, the deviations from the country mean of the fraction of the cash-flow rights held by th firm's 10 percent controlling shareholder (defined in Table I); (6) the interaction between CF rights and common law; and (7) the interactio between CF rights and anti-director rights. Table I contains definitions for the variables. Standard errors are shown in parentheses.

	Independent Variables							
Constant	Growth in Sales	Industry-adjusted GS	Common Law	Anti-director Rights	CF Rights	CF Rights * Common Law	CF Rights * Anti-director Rights	Overal R^2
Panel A: Tobin's q								
1.2986*	0.8275*		0.2441***					0.0735
(0.0836)	(0.1403)		(0.1400)					
1.2994*	0.8219*		0.2442***		0.2561***	−0.0510		0.0796
(0.0865)	(0.1403)		(0.1452)		(0.1345)	(0.2372)		
1.1559*	0.8314*			0.0735				0.0654
(0.1649)	(0.1403)			(0.0490)				
1.1562*	0.8296*			0.0735	0.4376***		−0.0671	0.0722
(0.1698)	(0.1400)			(0.0505)	(0.2675)		(0.0826)	
Panel B: Industry-adjusted Tobin's q								
0.0705		0.7505*	0.1982***					0.0659
(0.0704)		(0.1336)	(0.1199)					
0.0706		0.7506*	0.1981		0.2715**	−0.1401		0.0727
(0.0728)		(0.1337)	(0.1242)		(0.1275)	(0.2250)		
−0.0638		0.7534*		0.0659				0.0638
(0.1389)		(0.1336)		(0.0415)				
−0.0638		0.7527*		0.0659	0.4177***		−0.0649	0.0710
(0.1440)		(0.1333)		(0.0430)	(0.2536)		(0.0783)	

*Significant at 1 percent level.
**Significant at 5 percent level.
***Significant at 10 percent level.

Table V: Random Effects and Demeaned Ownership.

Table IV repeats the valuation tests with industry-adjusted q; Table V focuses on ownership deviations from country means.

13.4 Table VI: Ownership, Control, Wedge

Table VI

Ownership, Control, Wedge

Panel A classifies countries by legal origin and presents the average cash-flow rights, control rights, and the wedge (defined as the difference between control rights and cash flow rights). Table I defines the variables. Panel B reports tests of means for civil versus common law legal origin.

Country	N	CF Rights	Control Rights	Wedge
		Panel A: Means		
Argentina	20	0.38	0.48	0.10
Austria	20	0.47	0.56	0.10
Belgium	20	0.29	0.39	0.10
Denmark	20	0.30	0.41	0.10
Finland	20	0.30	0.38	0.08
France	20	0.23	0.37	0.13
Germany	20	0.30	0.37	0.07
Greece	20	0.48	0.52	0.04
Italy	20	0.35	0.51	0.16
Japan	20	0.25	0.26	0.01
Korea	20	0.18	0.24	0.06
Mexico	20	0.36	0.52	0.16
Netherlands	20	0.33	0.70	0.37
Norway	20	0.27	0.34	0.07
Portugal	20	0.46	0.49	0.03
Spain	20	0.26	0.33	0.07
Sweden	20	0.12	0.32	0.19
Switzerland	20	0.34	0.46	0.12
Civil law mean	**20**	**0.32**	**0.43**	**0.11**
Australia	20	0.25	0.30	0.05
Canada	20	0.25	0.41	0.17
Hong Kong	20	0.32	0.42	0.10
Ireland	20	0.29	0.30	0.01
Israel	19	0.24	0.40	0.16
New Zealand	20	0.24	0.33	0.08
Singapore	20	0.31	0.38	0.07
United Kingdom	20	0.14	0.25	0.10
United States	20	0.20	0.21	0.01
Common law mean	**20**	**0.25**	**0.33**	**0.08**
Sample mean	**20**	**0.29**	**0.39**	**0.10**
		Panel B: Test of Means (t-stats)		
Civil versus common law		1.9323**	2.1632*	0.90

*Significant at 5 percent level.
**Significant at 10 percent level.

Table VI: Ownership, Control, Wedge.

14 Key takeaways

14.1 Core mechanism

The three core predictions are: better investor protection raises q; higher controller cash-flow ownership raises q; and better investment opportunities raise q. The FOC $c_s(k, s^*) = 1 - \alpha$ is the central equation linking legal protection and ownership incentives to diversion.

14.2 For empirical corporate finance

The paper is a template for connecting a simple agency model to cross-country data. It also illustrates why one must distinguish cash-flow rights from control rights when studying firms with ultimate owners.

14.3 For research design

The identification is strongest for showing robust cross-sectional associations consistent with the model. It is less strong as a fully causal estimate because legal origin and ownership structures are not randomly assigned. A modern follow-up would seek shocks to investor protection, enforcement, or control rights to sharpen causal interpretation.